



二维坐标变换

$$A = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$$

$$\therefore A^T = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

$$AA^T = E$$

$$\therefore A_1 \cos\theta \cos\theta + \sin\theta \sin\theta$$

$$A_{12} = \cos\theta \cos\theta (-\sin\theta) + \cos\theta \sin\theta$$

$$A_{21} = -\sin\theta \cos\theta + \cos\theta \cos\theta$$

$$A_{22} = \sin\theta \sin\theta + \cos\theta \cos\theta$$

matlab坐标下的二维

$$T_{21/2r} = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} \quad \checkmark$$

